

Elevating PulseUp: Data Insights for a Healthier Bottom Line

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Executive Summary

Low conversion rates from free to premium customers, uneven engagement, and unclear results from recent marketing strategies are three main challenges that PulseUp, a freemium fitness company, is dealing with. To increase long-term revenue and user value, these problems must be resolved.

This project employs a data-driven methodology in five analytical domains: factor analysis to determine the primary perceptual drivers of user experience (trust, perceived value, and usability); customer segmentation to identify unique user profiles; linear regression to investigate the factors influencing spending among paying subscribers; a quasi-experimental design to assess the causal impact of the onboarding campaign; and logistic regression to construct a conversion prediction model.

Several findings stand out. Conversion increased by about 9.5 percentage points as a result of the onboarding campaign, which had a strong and statistically significant impact. Across all models, engagement (measured by session frequency and duration) consistently proved to be the best indicator of conversion. However, despite being helpful for segmentation, perceptual elements like perceived value and trust do not reflect how much consumers spend after converting, highlighting a crucial divergence between revenue drivers and conversion drivers.

Overall, these findings suggest that raising views on its own is not enough to increase revenue. In order to improve future decision-making, PulseUp should invest in greater behavioral data collecting and prioritize targeted engagement techniques and stronger monetization design, including pricing structure and upgrade options.

1. Factor Analysis & Survey Interpretation

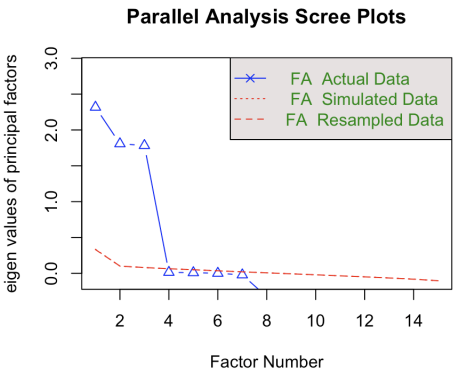
1.1 Justification For Using Factor Analysis

Factor analysis is an appropriate method for this survey because it seeks to measure latent psychological constructs (perceived value, usability, and trust) using multiple Likert-scale items. The Kaiser-Meyer-Olkin measure is 0.84, indicating good sampling adequacy. Additionally, Bartlett's Test of Sphericity is highly significant ($P < 0.001$), confirming that the variables are sufficiently correlated to justify factor analysis. Therefore, factor analysis is suitable for reducing the survey data into a smaller number of relevant constructs.

1.2 Number Of Factors

Based on several criteria, a **three-factor solution is kept**: the parallel analysis demonstrates that only the first three factors exceed simulated data, the scree plot shows a clear break after the third factor, and the first three eigenvalues are above 1.

Parallel analysis scree plot:



The graph supports the retention of three factors by graphically confirming the distinct decline in eigenvalues following the third factor.

1.3. Rotation Method

Tests were conducted for both orthogonal varimax and oblimin rotations. It appears that the latent constructs are independent because the correlations between the factors are close to zero. As a result, **varimax rotation is favored** since it provides a more straightforward and understandable structure while preserving data consistency.

1.4. Factor Interpretation

The factor structure is clear and well-defined. Factor 1, trust, is strongly loaded on ST1 - ST5, while Factor 2, perceived value, is strongly loaded on SV1 - SV5. Factor 3, usability, is strongly loaded on SU1 - SU5. All factor loadings exceed 0.5, and there are no significant cross-loadings. This indicates a clear and stable structure where each item measures a single underlying construct.

1.5. Model Fit

The model demonstrates a very good fit, with RMSEA of 0.005, TLI of 0.999, and RMSR of 0.01. These results indicate that the three-factor model effectively reproduces the data.

1.6. Reliability

Perceived value is 0.80, usability is 0.80, and trust is 0.81 according to Cronbach's alpha. All values exceed the standard threshold of 0.70, indicating good internal consistency for each construct.

1.7. Factor Scores

The average of the items related to each construct was used to calculate factor scores. Specifically, trust corresponds to the mean of ST1 - ST5, perceived value to the mean of SV1 - SV5, and usability to the mean of SU1 - SU5. At the individual level, this aggregate offers a concise representation of every latent construct. It makes it possible to place each responder on the three dimensions found in the factor analysis, facilitating their use in further analyses like segmentation, regression, and prediction.

1.8. Managerial Implications

The three factors (perceived value, usability, and trust) represent important drivers of PulseUp's customer experience. In practice, management can use these factors to guide specific decisions. For instance, perceived value can be used to optimize pricing and rewards strategies; usability can guide product improvements to increase engagement and retention; trust can be strengthened through transparent communication and data privacy policies, all of which can be used for customer segmentation. Management will be able to recognize various user profiles and modify marketing tactics appropriately. By tracking these variables over time and interpreting results such as conversion, engagement, and expenditure, they will enhance performance tracking and improve decision-making; resulting in more focused and effective marketing strategies.

2. Customer Segmentation

2.1 Assessment of the Need for Segmentation

Customers are heterogeneous, especially in the case of the Freemium subscription model, ranging from highly engaged users to passive trialists with low perceived value. One strategy with the same marketing message does not fit all. Segmentation will allow PulseUp to identify meaningful customer segments to personalise a new onboarding campaign, which maximises the conversion rate.

2.2 Variables and Data Preparation

We used a merged dataset combining behavioural metrics (*tenure_months, sessions_last_30d, avg_session_duration, emails_opened*) with perceptual factor scores (*score_value, score_usability, score_trust*).

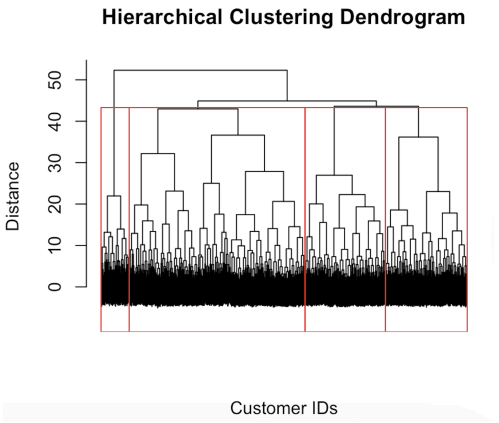
As these variables are measured on entirely different scales, all inputs were standardised using Z-score scaling:

$$z = (x - \mu) / \sigma$$

Without this adjustment, for example, the larger values in *avg_session_duration* would overshadow the smaller values in *score_trust,* leading to biased results.

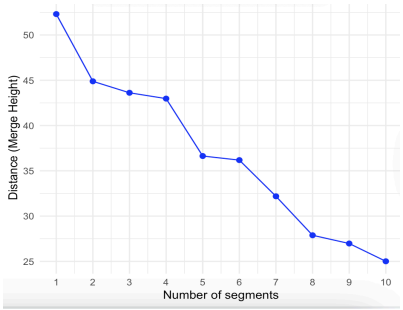
2.3 Clustering Methodology

We implemented a two-step cluster analysis to balance mathematical precision with practical managerial insights. First, we used Hierarchical Clustering to see how the data naturally branches out.
Dendrogram:



The Dendrogram visualised the hierarchical relationships, suggesting a natural and distinct cut at four main branches.

Elbow Plot:



The Elbow plot highlighted a clear inflexion point at four segments, confirming that introducing a fifth group would add unnecessary complexity without providing meaningful additional differentiation.

After confirming that a four-segment model offers the best compromise between statistical simplicity and strategic value, we applied the K-Means algorithm to classify each user into their respective group accurately.

2.4 Interpretation & profiling of segments

The four segments were profiled using the variables from the R output. The table below summarises the key behavioural and perceptual characteristics:

Variable	Seg 1 (n=394)	Seg 2 (n=919)	Seg 3 (n=1012)	Seg 4 (n=675)
BEHAVIORAL BASES (CLUSTERING VARIABLES)				
Avg. sessions (last 30d)	7.99	6.75	9.23	8.1
Avg. session duration (min)	25	20.4	29.5	24.1
Avg. value score (1–5)	3.97	3.96	3.91	4.18
Avg. trust score (1–5)	4.04	4.25	3.79	3.88
Conversion rate (%)	38.1	31.9	37.5	33.8
ADDITIONAL PROFILING VARIABLES				
Avg. spend (€/month)	9.05	8.05	9.77	8.59
% Female	50.5	47.8	50.6	48.9
% Male	49.5	52.2	49.4	51.1
Avg. age	35.6	35	35	34.5

Based on these profiles, the four segments can be meaningfully named and interpreted:

Segment 1 "Committed Converters" (13%) Moderate frequency, above-average session length, 38.1% conversion rate. Engaged and purposeful users with reasonably high trust.

Segment 2 "Trusting but Light Users" (31%) Highest trust score, but shortest sessions and lowest conversion rate. They believe in the app but don't use it intensively enough to feel the pull toward premium.

Segment 3 "Power Users with Friction" (34%) Most active users — highest sessions, longest duration, most spend (€9.77/month). But the lowest trust and perceived value act as a ceiling on conversion.

Segment 4 "High-Value Considerers" (23%) Highest value perception, but only 33.8% conversion. The gap likely stems from lower trust or price sensitivity.

2.5 Managerial interpretation & use of segments

The table below summarises targeting priority and the recommended marketing action per segment:

Segment	Key insight	Priority	Recommended action
Seg 3 — Power Users with Friction	Most active, highest spend, but lowest trust and value perception	High	Address trust barriers directly (data privacy messaging, social proof). Offer value-anchored premium trial. Highlight ROI of paid features to users who already use the app heavily.
Seg 4 — High-Value Considerers	Highest value perception, but conversion lags behind expectation	High	Reduce friction at conversion point (e.g., limited-time offer, one-click upgrade). These users already see the value — remove the final barrier. Test price anchoring and urgency nudges in onboarding.
Seg 1 — Committed Converters	Good conversion rate and engagement — the model target user	Medium	Protect and grow. Use personalised retention campaigns. Leverage as reference users for testimonials to other segments. Cross-sell premium feature bundles.
Seg 2 — Trusting but Light Users	High trust but low activity — under-engaged despite goodwill	Lower	Re-engagement campaign focused on habit formation (push notifications, usage streaks). Demonstrate concrete benefits of more frequent use. Long-term nurturing rather than immediate conversion push.

PulseUp should not communicate the same onboarding message to all users. Segments 3 and 4 represent the highest conversion opportunity; they account for over 1,600 users, but they need fundamentally different messages. Segment 3 needs trust-building; Segment 4 needs a nudge past the decision threshold.

3. Linear Regression Analysis

3.1 Managerial Question

The managerial question is: *What factors explain variation in monthly spending among paying PulseUp subscribers?*
We aim to assess whether perception-based factors (value, usability, trust) translate into higher revenue among paying users.
The dependent variable is **monthly_spend** (in euros). We analyze only converted users (monthly_spend > 0) in order to separate the decision to spend from the level of spending itself.
Linear regression might be appropriate since the dependent variable is continuous and allows us to estimate how customer characteristics, behavior, and perceptions are associated with spending.

3.2 Formula

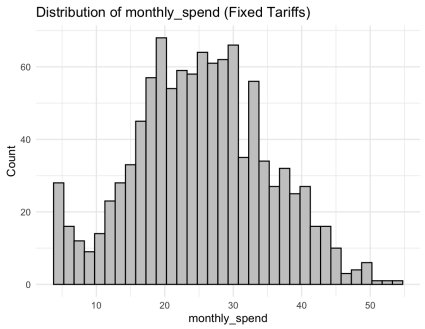
$monthly_spend_i = \beta 0 + \beta 1 \cdot Value_i + \beta 2 \cdot Usability_i + \beta 3 \cdot Trust_i + \beta 4 \cdot Sessions_i + \beta 5 \cdot Age_i + \epsilon i$
Where:

- Value, Usability, Trust: The latent constructs from our Factor Analysis (Part 1).
- Sessions: Number of app sessions in the last 30 days, representing engagement.
- Age: A demographic control variable.

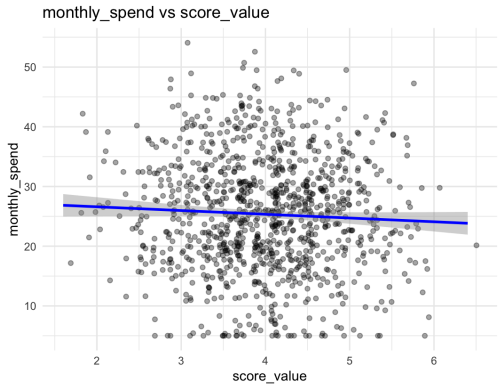
Other variables were excluded to avoid redundancy, as some metrics capture similar aspects of user behavior. The variable *trial_user* was omitted since the analysis focuses only on current paying subscribers, as mentioned earlier.

3.3 Visual Exploration of Spending Patterns

Before estimating the model, we examine the distribution of the dependent variable.



Monthly spending is relatively concentrated, with most observations in a narrow range. This suggests limited variation, which may reduce the ability to detect strong relationships.
We also examine the relationship between perceived value and spending:



The scatterplot shows no clear linear relationship. The fitted line is nearly flat, indicating only a weak association between perceived value and spending.

3.4 Regression Results and Interpretation

The regression results show that none of the explanatory variables are statistically significant at the 5% level.

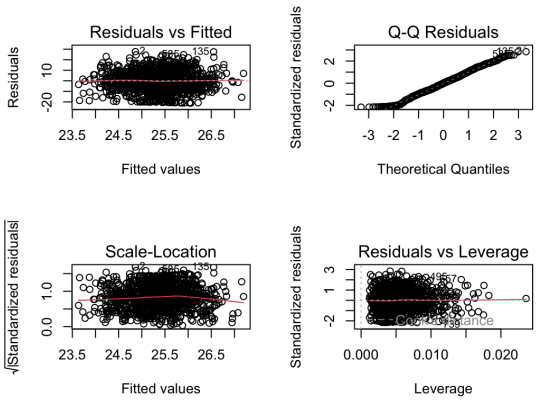
Value has a weak negative but insignificant effect, while Usability, Trust, Sessions and Age show minimal impact. Model fit is extremely low: $R^2 \approx 0.004$, and the F-test is not significant, indicating very limited explanatory power.

Taken together, these results suggest that the included variables do not meaningfully explain differences in spending among paying users.

3.5 Model Diagnostics

We also verified that the results are not driven by technical issues.

VIF values are close to 1, indicating no multicollinearity. Furthermore, the Breusch–Pagan test shows no heteroskedasticity, and robust standard errors confirm the findings.



The residual plots do not show strong patterns or major deviations from assumptions. There are no clear signs of model misspecification or influential outliers.

This confirms that the weak results are not due to model problems, but rather reflect a lack of explanatory power in the chosen variables.

3.6 Interpretation

The results suggest that monthly spending among paying users is not strongly driven by perception-based factors such as value, usability, or trust.

A possible explanation is that, once users convert, their spending becomes relatively stable. This implies that the drivers of conversion and the drivers of spending intensity are not the same.

3.7 Managerial Implications

Given the low explanatory power of the model, strong causal conclusions cannot be drawn.

From a management perspective, this suggests that improving perceptions of Value, Usability, or Trust is unlikely to boost ARPU among existing subscribers. This indicates that revenue growth may also depend on other levers, such as pricing structure, product packaging, or upgrade mechanisms.

Further analysis would be needed to provide more confident strategic conclusions and advice.

4. Experimentation / Quasi-Experiment

4.1 Causal Objective

We are trying to estimate the causal impact of the used campaign on user conversion. The campaign seeks to increase the conversion from free trial to paid subscription through new and redesign emails, personalized app messages, and more simplistic subscription prompts during the first days of using the app.

The variable that we are going to pay more attention to is the binary variable converted, is equal to 1 if the user subscribes and 0 if he is not. We are going to have a causal approach because the difference in conversion between exposed and non-exposed users may be affected by confounding factors, such as variations in the engagement of the users or time effects, rather than the effect of only the actual campaign.

4.2 The Limitations of Naive Comparisons

A simple interpretation is not enough to justify the causal approach. A before-and-after comparison is not accurate because conversion may change over time due to external factors, like seasonality or even the user composition. In other words, the changes after the campaign cannot be attributed only to the factor of the campaign.

A treated-control comparison is also biased because the campaign distribution wasn't random. In this scenario, the exposure to the campaign was guided by operational constraints. So, this selection bias can confound the relationship between treatment and conversion.

4.3 The Best Strategy: Difference-in-Difference

Knowing the biases that these other comparison methods can have, we use a Difference-in-Difference (DID) comparison, which compares changes in conversion between treated and control users:

$$(After - Before)_{Treated} - (After - Before)_{Control}$$

This method removes:

- Common time effects
- Time-invariant difference between groups

To be more specific this would be our model:

$$converted_i = \beta_0 + \beta_1 treatment_i + \beta_2 post_i + \beta_3 (treatment_i \times post_i) + \epsilon_i$$

Where:

- Treatment indicates people that were exposed to the campaign
- Post indicates the post-intervention period
- The interaction term captures the DID estimator

The coefficients are interpreted as:

- β_0: baseline conversion (control, before)
- β_1: pre-existing difference between groups
- β_2: common time effect
- β_3: causal effect of the campaign

4.4 Results and Validation through Group Means

The estimated interaction coefficient is $\beta_3 \approx 0.095$ and statistically significant ($p < 0.01$).

This results means that the onboarding campaign increased conversion by 9.5 percentage points for treated users compared to the control group.

Through R, we found that the DID estimate is consistent with group-level changes:

- Control: 0.320 -> 0.313 (−0.7 pp)
- Treated: 0.341 ->0.429 (+8.8 pp)

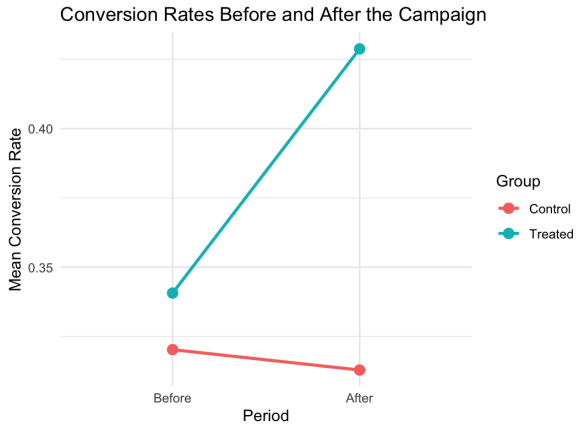
$$0.088 - (-0.007) = 0.095$$

This confirms the internal consistency of the estimation.

4.5 Robustness

Also, we decided to analyse the results, including the control variables, like age, tenure and engagement metrics. We saw that it doesn't change the estimated effect ($\beta_3 \approx 0.096$), indicating that the observed results are robust to observable differences among the different clients.
Among the chosen analysed controls, the engagement variables, like session frequency and duration, are strong predictors of conversion, highlighting the importance of the user activity in their different platforms.

4.6 Graphical illustration



4.7 Recommendations

To conclude, the results indicate that the campaign has a positive and apparently significant impact on conversion, with an increase of around 9.5% points.
We recommend Pulse Up to scale the campaign, as it could represent an important improvement in monetization. However, these results are dependent on the validity of the parallel trends assumption due to the not fully randomised treatment.

5. Logistic Regression

5.1 Managerial Problem & Justification

PulseUp's management wants to find out which users are most likely to upgrade from the free version to a paid subscription. This will help them to target marketing actions efficiently and maximise return on marketing investment (ROMI). The appropriate tool is logistic regression because the outcome variable, converted , is binary (1 = paid subscriber, 0 = free user), with an observed conversion rate of 35.0% (1,051 out of 3,000 users).

Unlike linear regression, which could predict probabilities outside the [0,1] range, logistic regression models the probability of conversion directly through a logit transformation, outputting values strictly between 0 and 1 for each user.

5.2 Model Specification

The following logistic regression model was estimated using maximum likelihood estimation:

$$\log(P / 1-P) = \beta_0 + \beta_1 \cdot \text{age} + \beta_2 \cdot \text{gender} + \beta_3 \cdot \text{tenure} + \beta_4 \cdot \text{sessions} + \beta_5 \cdot \text{duration} + \beta_6 \cdot \text{emails} + \beta_7 \cdot \text{score_value} + \beta_8 \cdot \text{score_usability} + \beta_9 \cdot \text{score_trust}$$

We included age and gender as demographic controls. We also included tenure as a measure of user maturity in the platform. We used sessions and average session duration as behavioral indicators of

frequency and depth of engagement. We finally used emails opened as a proxy for responsiveness to CRM activity. The three perceptual factor scores were included because perceived value, usability and trust are all potential motivators of subscription. We excluded the raw survey items in favour of factor scores to keep the model more concise and avoid redundancy across highly related items. Variables such as monthly spend were also excluded because they would overlap mechanically with conversion and the usefulness of the model for prediction would be reduced. This specification reflects a trade-off between parsimony and predictive performance, with the decision being to retain a limited set of interpretable predictors that management can act upon rather than maximising complexity.

Predictor variable	Coefficient	Standard Error	Z-value	P-value	Odds Ratio
Constant	-2.020	0.419	-4.821	<0.001	0.133
Age	0.001	0.004	0.309	0.757	1.001
Gender (Male)	0.033	0.077	0.426	0.670	1.033
Tenure (months)	0.005	0.003	1.566	0.117	1.005
Sessions last 30 days	0.034	0.014	2.496	0.013	1.035
ıverage session duration	0.025	0.005	5.225	<0.001	1.026
Emails opened	0.004	0.022	0.178	0.859	1.004
Perceived value score	-0.019	0.049	-0.378	0.705	0.982
Usability score	0.014	0.050	0.274	0.784	1.014
Trust score	0.094	0.050	1.888	0.059	1.099

5.3 Interpretation of Key Coefficients

Log odds are not directly intuitive. The coefficients were interpreted using odds ratios. The results show that behavioural engagement is the strongest predictor of conversion: an additional session in the last 30 days increases the odds of conversion by around 3.5% (Odds ratio = 1.035, p = 0.013), and an extra minute of average session duration increases them by around 2.6% (Odds ratio = 1.026, p < 0.001). This suggests that users who return to the app more often and spend more time on it are much more likely to become paying subscribers. Trust also has a positive effect (Odds ratio = 1.099), but this is only marginally significant (p = 0.059) and therefore the evidence is weaker. On the other hand, age, gender, how long someone has been with the company, how often emails are opened, perceived value and usability do not have a statistically significant effect in this model.

5.4 Model Fit Evaluation

```
  customer_id converted pred_prob pred_class
1           1         0 0.3384468          0
2           2         0 0.3877273          0
3           3         1 0.3112562          0
4           4         0 0.3937039          0
5           5         0 0.3649406          0
6           6         0 0.4053087          0
```

```
          Predicted
Real     0    1
   0  1939   10
   1  1037   14
```

The model can identify non-converters but struggles significantly when it comes to identifying actual converters. The pseudo-R^2 of 1.0% suggests that the predictors offer a very limited explanation of the variation in conversion. The confusion matrix reveals the core problem: at the default 0.5 threshold,

the model classifies almost everyone as non-converting (only 24 users it predicts as 1), catching just 14 out of 1,051 true converters — a sensitivity of 1.3%.

This outcome is diagnostically important: conversion at PulseUp is not strongly predicted by the variables available in this dataset, suggesting that key drivers may be missing.

5.5 Managerial Recommendations

Although the model is ineffective as a strict classifier, it is still valuable as a ranking tool. PulseUp should not use it to determine whether a user will convert. Management should use the predicted probabilities to prioritise users who are most likely to convert. The code already identifies the top non-converted users with the highest predicted probabilities.

High-probability non-converted users should be prioritised for premium conversion campaigns by PulseUp, because stronger behavioural engagement is already displayed by them. Usage frequency and depth are the main predictive drivers, so management should focus on strengthening onboarding and in-app journeys to increase repeat sessions and time spent in the app. The weak model fit indicates that important drivers of conversion are still missing from the dataset, so future data collection should include richer behavioural signals such as workout completion, feature usage, notification clicks, or payment-friction variables.

Conclusion

Overall, the analysis gives a clear and structured view of how users behave in PulseUp, but also shows that different factors matter at different stages of the funnel. The factor analysis and segmentation confirm that trust, perceived value, and usability are relevant for understanding how users perceive the product and why they convert. However, the regression results suggest that these same factors do not explain how much users spend after converting. This highlights an important distinction between drivers of conversion and drivers of revenue.

At the same time, the most consistent insight across models is the role of engagement. Both the logistic regression and the experiment show that users who are more active are more likely to convert, and the onboarding campaign had a meaningful impact, increasing conversion by around 9.5 percentage points. However, the weak predictive power of the models suggests that important variables are missing, so the results should be interpreted with caution.

Overall, this suggests that improving perceptions alone is not enough to drive revenue growth. PulseUp should combine targeted onboarding and engagement strategies with a stronger focus on monetization design, such as pricing and upgrade mechanisms. At the same time, collecting richer behavioral data would help better understand what actually drives user decisions.

Appendix

Part 1 - Codes

```
customer_survey <- read.csv("pulseup_prepared.csv")

library(dplyr)
library(psych)
```

```
# 1) Assess whether factor analysis is appropriate

items <- customer_survey %>%
  select(
    SV1, SV2, SV3, SV4, SV5,
    SU1, SU2, SU3, SU4, SU5,
    ST1, ST2, ST3, ST4, ST5
  ) %>%
  na.omit()

cor_mat <- cor(items)

kmo_result <- KMO(cor_mat)
print(kmo_result)

bartlett_result <- cortest.bartlett(cor_mat, n = nrow(items))
print(bartlett_result)
```

```
# 2) Decide on the number of factors

eigen_values <- eigen(cor_mat)$values
print(eigen_values)

fa.parallel(items, fa = "fa", fm = "ml", n.iter = 100)
```

```
# 3) Perform exploratory factor analysis

efa_oblimin <- fa(
  items,
  nfactors = 3,
  rotate = "oblimin",
  fm = "ml",
  scores = "regression"
)

print(efa_oblimin)
print(efa_oblimin$loadings, cutoff = 0.30)

efa_varimax <- fa(
  items,
  nfactors = 3,
  rotate = "varimax",
  fm = "ml",
  scores = "regression"
)

print(efa_varimax)
print(efa_varimax$loadings, cutoff = 0.30)
```

```
# 4) Decide on the rotation method

print(efa_oblimin$Phi)

# If factors are correlated, oblimin is more appropriate
# If factors are weakly correlated, varimax can also be considered
```

```r
# 5) Interpret and label the factors

print(efa_oblimin$loadings, cutoff = 0.30)
print(efa_varimax$loadings, cutoff = 0.30)
```

```r
# 6) Assess reliability

alpha_value <- alpha(customer_survey %>% select(SV1, SV2, SV3, SV4, SV5))
alpha_usability <- alpha(customer_survey %>% select(SU1, SU2, SU3, SU4, SU5))
alpha_trust <- alpha(customer_survey %>% select(ST1, ST2, ST3, ST4, ST5))

print(alpha_value$total)
print(alpha_usability$total)
print(alpha_trust$total)
```

```r
# 7) Compute factor scores

survey_scores <- customer_survey %>%
  mutate(
    perceived_value = rowMeans(select(., SV1, SV2, SV3, SV4, SV5), na.rm = TRUE),
    usability = rowMeans(select(., SU1, SU2, SU3, SU4, SU5), na.rm = TRUE),
    trust = rowMeans(select(., ST1, ST2, ST3, ST4, ST5), na.rm = TRUE)
  ) %>%
  select(customer_id, perceived_value, usability, trust)

head(survey_scores)
```

Part 2 - Codes

```r
## Load required libraries and set seed
packages <- c("psych", "dplyr", "cluster", "ggplot2", "ggdendro")
installed_packages <- packages %in% installed.packages()[,"Package"]
if (any(installed_packages == FALSE)) {
  install.packages(packages[!installed_packages])
}
lapply(packages, library, character.only = TRUE)

set.seed(1)

# Import Data
factor_data <- read.csv(file.choose())
behavior_data <- read.csv(file.choose())

# Clean Data
factor_data_clean <- factor_data %>%
  select(customer_id, SV1:score_trust)

# Merge Data
pulseup_data <- merge(behavior_data, factor_data_clean, by = "customer_id")
describe(pulseup_data)
pulseup_data <- pulseup_data %>% select(-ends_with(".y"))
names(pulseup_data) <- gsub("\\.x$", "", names(pulseup_data))
describe(pulseup_data)

# Clustering variables
clustering_vars <- pulseup_data %>%
  select(tenure_months, sessions_last_30d, avg_session_duration, emails_opened,
         score_value, score_usability, score_trust)
scaled_data <- scale(clustering_vars)
summary(scaled_data)
```

```r
dist_matrix <- dist(scaled_data, method = "euclidean")

hc_model <- hclust(dist_matrix, method = "ward.D2")

#Plot the dendrogram
plot(hc_model,
     main = "Hierarchical Clustering Dendrogram",
     xlab = "Customer IDs",
     ylab = "Distance",
     labels = FALSE)
rect.hclust(hc_model, k = 4, border = "red")


# Elbow plot for first 10 segments

sort_height <- sort(hc_model$height, decreasing = TRUE)
x <- 1:10
y <- sort_height[1:10]

ggplot(data = data.frame(x, y), aes(x = x, y = y)) +
  geom_line(color = "blue") +
  geom_point(color = "blue", size = 2) +
  scale_x_continuous(breaks = seq(1, 10, 1), labels = seq(1, 10, 1)) +
  labs(x = "Number of segments",
       y = "Distance (Merge Height)",
       title = "Elbow Plot using Hierarchical Fusion Heights") +
  theme_minimal()

SEG_standardized <- as.data.frame(scaled_data)
```

```r
# Run k-means with 4 segments: we obtain the number of segment each member belongs to
set.seed(1)
seg_kmeans <- kmeans(x = data.frame(SEG_standardized$tenure_months,
                                    SEG_standardized$sessions_last_30d,
                                    SEG_standardized$avg_session_duration,
                                    SEG_standardized$emails_opened,
                                    SEG_standardized$score_value,
                                    SEG_standardized$score_usability,
                                    SEG_standardized$score_trust),
                     centers = 4, nstart = 25)
# Adding the segment number to the main database
pulseup_data$segment <- as.factor(seg_kmeans$cluster)

# Check the number of clients in each segment
table(pulseup_data$segment)

segment_column <- seg_kmeans$cluster
SEG_standardized_with_segments <- cbind(SEG_standardized, segment = segment_column)

write.csv(SEG_standardized_with_segments, file = "pulseup_segmentation_results.csv", row.names = TRUE)

library(dplyr)
library(tidyr)

final_interpretation <- pulseup_data %>%
  group_by(segment) %>%
  summarise(
    n = n(),
    # Key Segmentation Variables
    Avg_Sessions = mean(sessions_last_30d),
    Avg_Trust = mean(score_trust),
    Avg_Value = mean(score_value),
```

Part 3 - Codes

```r
# PART 3 – Linear Regression

# ============================================================
# Objective: Identify drivers of monthly spending among paying users.
rm(list=ls()); gc()
# 1) Libraries
library(dplyr)
library(ggplot2)
library(car)       # For VIF analysis
library(lmtest)    # For Breusch-Pagan test
library(sandwich)  # For HC1 robust standard errors
library(lm.beta)   # For standardized coefficients
# 2) Data Preparation
# Loading the prepared dataset
pulseup <- read.csv(file.choose())
# Managerial restriction: analyzing spending intensity among active payers
# We focus on users who are already contributing to revenue.
spending_data <- pulseup %>%
  filter(monthly_spend > 0)
```

```r
# ============================================================================
# 1) Visual check: Examining functional forms
# ============================================================================
# Histogram to identify fixed pricing tiers (spikes in distribution)
ggplot(spending_data, aes(x = monthly_spend)) +
  geom_histogram(binwidth = 1.5, fill = "grey80", color = "black") +
  labs(title = "Distribution of monthly_spend (Fixed Tariffs)",
       x = "monthly_spend", y = "Count") +
  theme_minimal()

# Jittered scatter plot to visualize overlapping data points on pricing tiers
ggplot(spending_data, aes(x = score_value, y = monthly_spend)) +
  geom_jitter(alpha = 0.35, width = 0.15) +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  labs(title = "monthly_spend vs score_value",
       x = "score_value", y = "monthly_spend") +
  theme_minimal()

# ============================================================================
# 2) Model Estimation
# ============================================================================
# Estimating the impact of perceptions, behavior, and demographics
m1 <- lm(monthly_spend ~ score_value + score_usability + score_trust +
           sessions_last_30d + age, data = spending_data)

cat("\n--- OLS Regression Summary ---\n")
print(summary(m1))

#
```

```r

# ============================================================================
# 3) Assessing collinearity (VIF)
# ============================================================================
cat("\n--- VIF for collinearity check ---\n")
print(vif(m1)) # Checking if survey constructs are redundant (Goal: VIF < 5)

# ============================================================================
# 4) Heteroskedasticity test + robust SEs
# ============================================================================
cat("\n--- Breusch-Pagan test ---\n")
print(bptest(m1)) # Testing for non-constant variance

# Robust Coefficients (HC1) to account for fixed pricing variance
cat("\n--- Coefficients with robust SE (HC1) ---\n")
print(coeftest(m1, vcov = vcovHC(m1, type = "HC1")))

# ============================================================================
# 5) Managerial translation
# ============================================================================
# Standardized coefficients to determine relative driver importance
cat("\n--- Standardized coefficients (Beta) ---\n")
print(round(summary(lm.beta(m1))$coefficients, 4))

# ============================================================================
# 6) Residual diagnostics
# ============================================================================
# Visual audit of model assumptions (linearity, normality, outliers)
par(mfrow=c(2,2))
plot(m1)
```

38:1 (Untitled) R Script

Part 4 - Codes

```r
library(dplyr)
library(psych)
library(lmtest)
library(sandwich)
library(ggplot2)

customer_survey <- read.csv("pulseup_prepared.csv")

#1) Assess whether factor analysis is appropriate
items <- customer_survey %>%
  select(SV1, SV2, SV3, SV4, SV5,
         SU1, SU2, SU3, SU4, SU5,
         ST1, ST2, ST3, ST4, ST5)

cor_mat <- cor(items)

kmo_result <- KMO(cor_mat)
print(kmo_result)

bartlett_result <- cortest.bartlett(cor_mat, n = nrow(items))
print(bartlett_result)

#2) Decide on the number of factors
eigen_values <- eigen(cor_mat)$values
print(eigen_values)

fa.parallel(items, fa = "fa", fm = "ml", n.iter = 100)

#3) Perform exploratory factor analysis
efa_oblimin <- fa(items,
                  nfactors = 3,
                  rotate = "oblimin",
                  fm = "ml",
                  scores = "regression")

print(efa_oblimin)
print(efa_oblimin$loadings, cutoff = 0.30)

efa_varimax <- fa(items,
                  nfactors = 3,
                  rotate = "varimax",
                  fm = "ml",
                  scores = "regression")

print(efa_varimax)
print(efa_varimax$loadings, cutoff = 0.30)
```

```r
#4) Decide on the rotation method
efa_oblimin <- fa(items,
                  nfactors = 3,
                  rotate = "oblimin",
                  fm = "ml",
                  scores = "regression")

print(efa_oblimin$Phi)

efa_varimax <- fa(items,
                  nfactors = 3,
                  rotate = "varimax",
                  fm = "ml",
                  scores = "regression")

#5) Interpret and label the factors
print(efa_oblimin$loadings, cutoff = 0.30)
print(efa_varimax$loadings, cutoff = 0.30)

#6) Assess reliability
alpha_value <- alpha(customer_survey %>% select(SV1, SV2, SV3, SV4, SV5))
alpha_usability <- alpha(customer_survey %>% select(SU1, SU2, SU3, SU4, SU5))
alpha_trust <- alpha(customer_survey %>% select(ST1, ST2, ST3, ST4, ST5))

print(alpha_value$total)
print(alpha_usability$total)
print(alpha_trust$total)

#7) Compute factor scores and explain how you created them
survey_scores <- customer_survey %>%
  mutate(
    perceived_value = rowMeans(select(., SV1, SV2, SV3, SV4, SV5)),
    usability = rowMeans(select(., SU1, SU2, SU3, SU4, SU5)),
    trust = rowMeans(select(., ST1, ST2, ST3, ST4, ST5))
  ) %>%
  select(customer_id, perceived_value, usability, trust)

head(survey_scores)

#8) Prepare the data for DID
did_data <- customer_survey %>%
  mutate(
    converted = as.numeric(converted),
    treatment = as.numeric(treatment),
    post = as.numeric(post)
  )
```

```r
#9) Create the interaction term
did_data <- did_data %>%
  mutate(treat_post = treatment * post)

head(did_data %>% select(customer_id, converted, treatment, post, treat_post))

#10) Estimate the DID model
did_model <- lm(converted ~ treatment + post + treat_post, data = did_data)

summary(did_model)

#11) Robust standard errors
coeftest(did_model, vcov = vcovHC(did_model, type = "HC1"))

#12) DID model with controls
did_model_controls <- lm(
  converted ~ treatment + post + treat_post +
    age + tenure_months + sessions_last_30d +
    avg_session_duration + emails_opened,
  data = did_data
)

summary(did_model_controls)

coeftest(did_model_controls, vcov = vcovHC(did_model_controls, type = "HC1"))

#13) Compute average conversion by group and period
did_means <- did_data %>%
  group_by(treatment, post) %>%
  summarise(
    mean_conversion = mean(converted, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

print(did_means)
```

```r
#14) Manual DID estimate
control_before <- did_means %>%
  filter(treatment == 0, post == 0) %>%
  pull(mean_conversion)

control_after <- did_means %>%
  filter(treatment == 0, post == 1) %>%
  pull(mean_conversion)

treated_before <- did_means %>%
  filter(treatment == 1, post == 0) %>%
  pull(mean_conversion)

treated_after <- did_means %>%
  filter(treatment == 1, post == 1) %>%
  pull(mean_conversion)

did_estimate_manual <- (treated_after - treated_before) - (control_after - control_before)
print(did_estimate_manual)

#15) Plot average conversion before and after by group
did_plot_data <- did_means %>%
  mutate(
    group = ifelse(treatment == 1, "Treated", "Control"),
    period = ifelse(post == 1, "After", "Before"),
    period = factor(period, levels = c("Before", "After"))
  )

ggplot(did_plot_data,
       aes(x = period, y = mean_conversion, group = group, color = group)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Conversion Rates Before and After the Campaign",
    x = "Period",
    y = "Mean Conversion Rate",
    color = "Group"
  ) +
  theme_minimal()
```

Part 5 - Codes

```r
library(dplyr)
library(pROC)
library(caret)

# ── Prepare the dataset for logistic regression ──────────────────────
LOG <- customer_survey %>%
  select(converted, age, gender, tenure_months,
         sessions_last_30d, avg_session_duration,
         emails_opened, score_value, score_usability, score_trust)

# Descriptive statistics
summary(LOG)
str(LOG)
head(LOG, 10)
sapply(LOG, function(x) sum(is.na(x)))  # Should all be 0

#1) State the managerial problem and justify logistic regression
# ── PulseUp wants to predict which users will convert to a paid subscription
# ── Outcome: converted (binary: 1 = paid subscriber, 0 = not converted)
# ── Logistic regression is appropriate because:
#    - The outcome is binary (not continuous)
#    - We want probabilities between 0 and 1
#    - Linear regression could predict values outside [0,1]

# Check class balance
table(customer_survey$converted)
prop.table(table(customer_survey$converted))
# ~35% conversion rate → reasonably balanced, no severe imbalance issue

#2) Specify and estimate the logistic regression model
model_logit <- glm(converted ~ age + gender + tenure_months +
                     sessions_last_30d + avg_session_duration +
                     emails_opened + score_value + score_usability + score_trust,
                   data = customer_survey,
                   family = binomial)

print(summary(model_logit))


#3) Interpret coefficients — odds ratios (not raw log-odds)
odds_ratios <- exp(cbind(OR = coef(model_logit), confint(model_logit)))
print(odds_ratios)


# OR > 1 → variable increases the odds of conversion
# OR < 1 → variable decreases the odds of conversion
# OR = 1 → no effect

#4) Assess overall model fit — Pseudo R² (McFadden)
# ── As shown in class: Pseudo R² = 1 - LL(model)/LL(null)
null_model <- glm(converted ~ 1,
                  data = customer_survey,
                  family = binomial)

pseudo_r2 <- 1 - logLik(model_logit) / logLik(null_model)
print(pseudo_r2)
# Interpretation:
# 0.1–0.2 = acceptable fit
# 0.2–0.4 = good fit (McFadden scale is different from OLS R²)
```

```r
#5) Compute predicted probabilities
customer_survey$pred_prob  <- predict(model_logit, type = "response")
customer_survey$pred_class <- ifelse(customer_survey$pred_prob > 0.5, 1, 0)

head(customer_survey %>% select(customer_id, converted,
                                pred_prob, pred_class))

#6) Evaluate model performance — Confusion Matrix
confusion <- table(Real = customer_survey$converted,
                   Predicted = customer_survey$pred_class)
print(confusion)

# Manually compute key metrics
TP <- confusion[2, 2]
TN <- confusion[1, 1]
FP <- confusion[1, 2]
FN <- confusion[2, 1]

accuracy    <- (TP + TN) / sum(confusion)
sensitivity <- TP / (TP + FN)   # True Positive Rate (hit rate)
specificity <- TN / (TN + FP)   # True Negative Rate

cat("Accuracy:   ", round(accuracy, 3), "\n")
cat("Sensitivity:", round(sensitivity, 3), "\n")
cat("Specificity:", round(specificity, 3), "\n")

#7) Managerial use — identify top unconverted users to target
top_targets <- customer_survey %>%
  filter(converted == 0) %>%
  arrange(desc(pred_prob)) %>%
  select(customer_id, pred_prob, sessions_last_30d,
         score_value, score_trust, emails_opened)

head(top_targets, 20)
# These are non-converted users with the highest probability of converting
# → Priority targets for PulseUp's next marketing push
```